MIL-E-1/1056D
23 March 1971
SUPERSEDING
MIL-E-1/1056C
31 December 1964

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 12SP7D

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Magnetic deflection and focus

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ehk	Rg1	Alt
Unit:	V	Vdc	Vdc	kVdc	Vdc	Meg	ft
Maximum:	6.9	0, -180	770	15	-125	Note 1	60,000
Minimum:	5.7	---	---	4	---	---	---
TEST CONDITIONS:	6.3	Adjust	250	9	---	---	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
Ⓓ 1216	Base material insulating quality		---	---	---	---	---	---
1002	Barometric pressure, reduced	54 mmHg	---	---	---	---	---	---
5216	Cathode illumination		---	---	---	---	---	---
1331	Direct-interelectrode capacitance	Ck to all	---	---	Ck	---	8	pF
		Cg1 to all	---	---	Cg1	---	10	pF
		Cg2 to all	---	---	Cg2	---	10	pF
1141	Pressure (implosion)		---	---	---	---	---	---
5111	Vibration (cathode-ray tubes)		---	---	Width	---	2	mm
	Quality conformance inspection, part 1							
5106	Bulb, screen, and faceplate quality		---	---	---	---	---	---
1301	Heater current		---	---	If	540	660	mA
5201	Voltage breakdown		---	---	---	---	---	---
5201	Voltage breakdown (magnetic types)		---	---	---	---	---	---
5206	Gas ratio	See note 2	---	---	Gr	---	0.25	---
---	Neck straightness	See note 3	---	---	---	---	---	---
---	Cathode image	See note 4	---	---	---	---	20	%
5216	Stray light emission (conventional types)	Eb = 15,000 Vdc; Ec2 = 770 Vdc	---	---	---	---	---	---
5231	Spot position (magnetic deflection)	See note 5	---	---	---	---	20	mm
5236	Zero-bias anode current (magnetic deflection)		---	---	---	---	---	---
5241	Grid cutoff voltage		---	---	Ecо	-27	-63	Vdc
5251	Grid No. 1 leakage current		---	---	---	---	---	---
5251	Grid No. 2 leakage current		---	---	---	---	---	---
---	Aperture alinement	See note 6	---	---	Distance	---	8	mm
	Quality conformance inspection, part 2							
5201	Electrode current (grid No. 2 current)	Ec1 = 0	---	---	Ic2	---	50	μAdc
5101	Side terminal and base alinement	Pin No. 3	---	---	---	---	---	---
Ⓓ 1111	Base pin solder depth		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 2 -Continued							
5101	Neck and bulb alinement (magnetic types)		---	---	---	---	0.25	inch
5101	Face tilt		---	---	---	---	---	---
5223	Modulation	Ib = 200 μAdc	---	---	ΔEc1	---	35	Vdc
5221	Screen		---	---	---	---	---	---
5226	Line width A (magnetic deflection)	Ib = 200 μAdc	---	---	Width	---	0.38	mm
5226	Line width C (magnetic deflection)	Ib = 200 μAdc	---	---	Width	---	0.60	mm
5246	Focusing ampere turns	Ib - 200 μAdc; D - 3.25 inches	---	---	AT	450	550	---
5251	Heater-cathode leakage current		---	---	---	---	---	---
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C: t = 500 hours (min); Eb = 13,200 Vdc; Ec2 = 770 Vdc; Ib = 60 μAdc	---	---	---	---	---	---
---	Life-test end points	Ib = 200 μAdc;						
		Line width A	---	---	Width	---	0.38	mm
		Line width B	---	---	Width	---	0.60	mm
		Modulation	---	---	ΔEc	---	35	Vdc
		Heater-cathode leakage current	---	---	---	---	---	---
		Grid 1 leakage current	---	---	---	---	---	---
		Grid 2 leakage current	---	---	---	---	---	---
		Stray light emission (conventional types)	---	---	---	---	---	---

NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 Meg.
2. This test to be performed at the conclusion of the holding period.
3. The neck and base straightness shall be determined by the insertion of the tube neck in a cylinder 5-inches long and 1.5-inches maximum inside diameter. This cylinder shall move freely between the reference line and the base of the assembled tube.

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NOTES: -Continued

4. The test conditions shall be Eb = 7 kV, Ec2 = 250, focus coil current = 0, Ec1 shall be varied until the focused image of the cathode is obtained. The percentage of dead area shall be that portion of the image which is black in respect to the general brilliant area.

5. There shall be no displacement due to hand capacitance or external electrostatic field.

6. The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified herein. Ec1 shall not be held at zero for more than approximately 30 seconds to prevent damage to the screen.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH
Air Force - 19

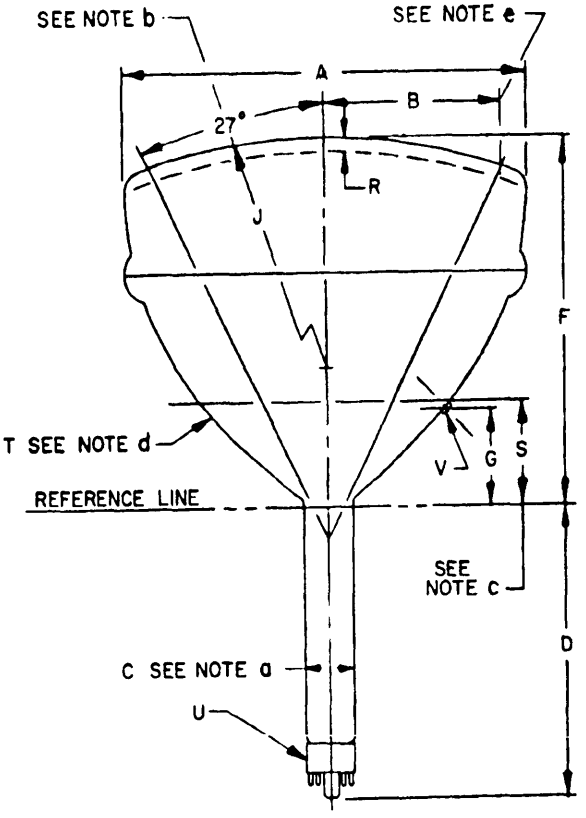
Preparing activity: Navy - EC

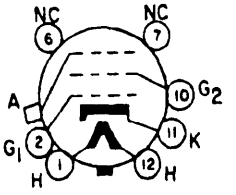
Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	12.340(313.44)	12.500(317.50)
B	5.500(139.70)	
C	1.384 (35.16)	1.500 (38.10)
D	8.060(204.72)	8.440(214.38)
F	10.310(261.87)	10.690(271.53
G	2.630 (66.80)	3.380 (85.85)
R	.300 (7.62)	.400 (10.16)
S	2.970 (75.44)	3.090 (78.49)
Quality conformance inspection, part 3 (periodic check)		
T	Bulb: J-99 1/2 A1, clear or filter glass	
U	Base: B7-51 or B5-57	
V	Contact: J1-21	
Reference dimensions (see note f)		
J	40.000 R (1016.00)	



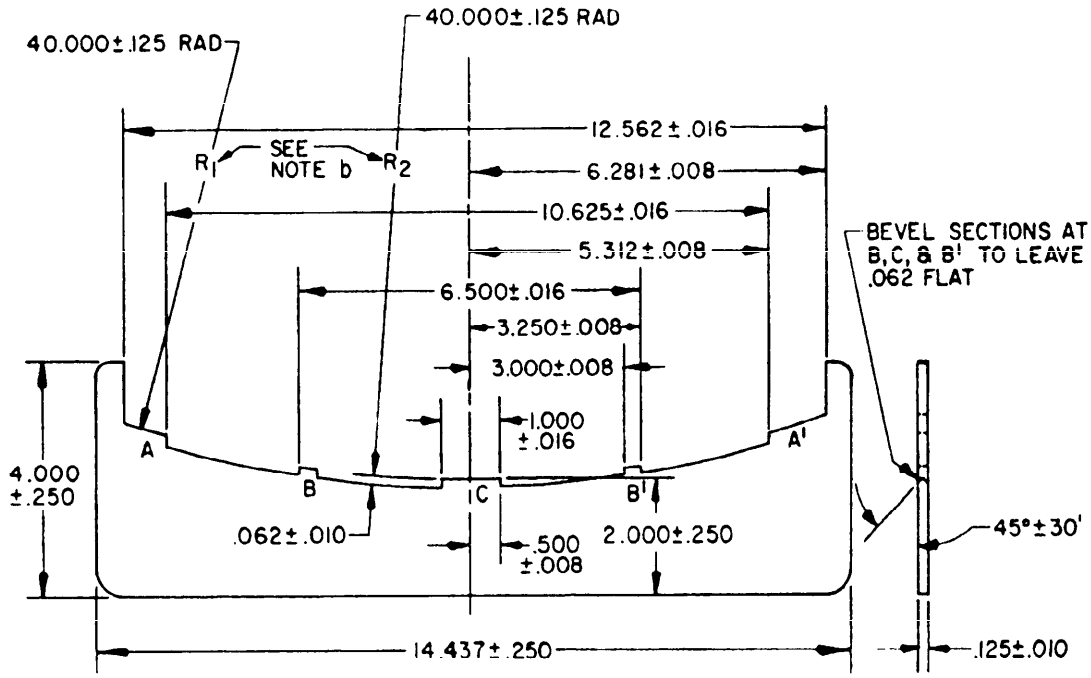


PIN CONNECTIONS

NOTES:
a. Dimension C applies to the entire length of neck below reference line. Reference line shall be determined by position where gage 1.500(38.10 mm)±.003(.08 mm) inches I.D. and 2.000(50.80 mm) inches long will rest on bulb cone.
b. To meet the requirements of faceplate gage as shown on figure 2.
c. Line established by ring gage of I.D. 7.188(182.58 mm)±.016(.41 mm) inches.
d. To meet the requirements of bulb contour gage as shown on figure 3.
e. Useful screen radius.
f. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 12SP7D.

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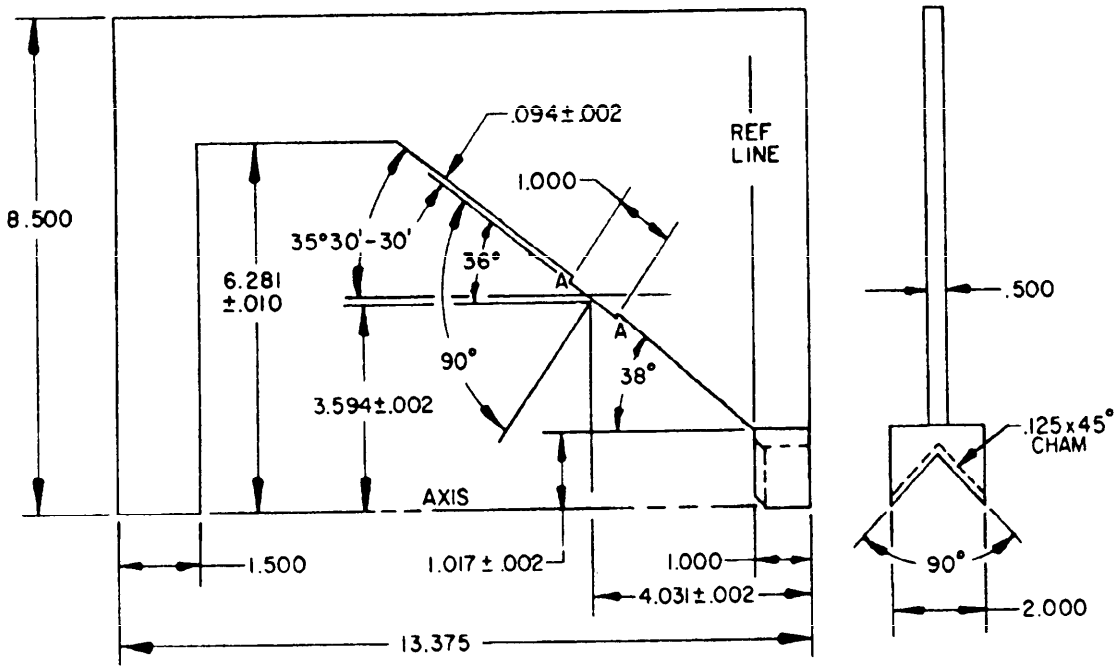


INCHES	MM	INCHES	MM	INCHES	MM
.008	.20	1.000	25.40	6.281	159.64
.010	.25	2.000	50.80	6.500	165.10
.016	.41	3.000	76.20	10.625	269.88
.062	1.57	3.250	82.55	12.562	319.07
.125	3.18	4.000	101.60	14.437	366.07
.250	6.35	5.312	134.92	40.000	1016.00
.500	12.70				

NOTES:
a. All dimensions in inches.
b. R2 shall be equal to the absolute value of R1 ±.050(1.27 mm) ±.002(.05 mm).
c. Instructions for use: Contour of face shall be checked with gage at two orientations approximately 90 degrees. Reject if gap between bulb face and gage points B, B', or C is greater than .100(2.54 mm). Reject if bulb does not seat on steps A and A'.

FIGURE 2. Faceplate gage.

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INCHES	MM	INCHES	MM
.002	.05	1.500	38.10
.010	.25	2.000	50.80
.094	2.39	3.594	91.29
.125	3.18	4.031	102.39
.500	12.70	6.281	159.60
1.000	25.40	8.500	215.90
1.017	25.83	13.375	339.73

NOTES:
a. All dimensions in inches.
b. Tolerances shall be ±.016(4.06 mm) on fractions, ±.005(.13 mm) on decimals, ±1/2° on angles, unless otherwise specified.
c. Instructions for use: Reject if bulb does not rest on flat 'A-A' when bulb neck is seated in V-block.

FIGURE 3. Bulb contour gage.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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